WoSM Academy

MAT231-F27 · CALCULUS II

Unconstrained optimisation

Critical points, the second-derivative test, convexity

Dr. Ali Khudiyev
WoSM Academy

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Last time

- The Hessian H_f: the symmetric matrix of second partial derivatives.
- **Taylor:** $f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2}\mathbf{h}^\top H_f(\mathbf{a})\mathbf{h} + o(\|\mathbf{h}\|^2)$.
- The quadratic term is a quadratic form: bowl (positive definite), cap (negative definite) or saddle (indefinite).

Today we use this to find and classify the highest and lowest points of a function of several variables — the problem behind every fit", train" and design".

This lecture

01 Critical points

02 The second-derivative test

03 Global extrema

04 Convexity

05 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Section 4.7; for convexity, Boyd & Vandenberghe, *Convex Optimization*, Section 3.1.

01

Critical points

Local and global extrema

Let $f : U \to \mathbb{R}$ with $U \subseteq \mathbb{R}^n$, and $\mathbf{a} \in U$.

Definition

- $\mathbf{a}$ is a **local minimum** of f if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in U$ in some ball $B(\mathbf{a}, r)$.
- $\mathbf{a}$ is a **global minimum** if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in U$.
- **Strict**: $<$ instead of $\leq$, for $\mathbf{x} \neq \mathbf{a}$. Maxima: reverse the inequalities.

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- Extrema = minima and maxima. Maximising f is minimising $-f$, so we usually speak of minima.
- Fitting a model, training a network, choosing a design: each minimises a function of many variables (a loss, an error, a cost).

Fermat's theorem

Theorem (Fermat)

If f has a local extremum at an interior point $\mathbf{a}$ of U and is differentiable at $\mathbf{a}$, then $\nabla f(\mathbf{a}) = \mathbf{0}$.

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Proof. For each i, the function $g(t) = f(\mathbf{a} + t\mathbf{e}_i)$ of one variable has a local extremum at $t = 0$, so $0 = g'(0) = f_{x_i}(\mathbf{a})$ (Calculus I). □

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Definition

A **critical point** of f is an interior point $\mathbf{a}$ with $\nabla f(\mathbf{a}) = \mathbf{0}$: the tangent plane there is horizontal. A critical point that is not a local extremum is a **saddle point**.

Points where f is not differentiable, and boundary points, are checked separately.

The converse is false

Example ($f(x,y) = x^2 - y^2$)

$\nabla f = (2x, -2y)$ vanishes at $(0,0)$. Along the x-axis $f = x^2$ has a minimum there, along the y-axis $f = -y^2$ a maximum. A saddle.

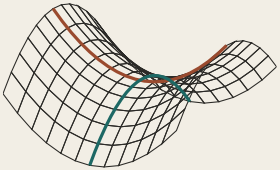
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Example ($f(x, y) = x^3 + y^2$)

$\nabla f = (3x^2, 2y)$ vanishes at $(0, 0)$, but $f(t, 0) = t^3$ takes both signs near $t = 0$. A saddle again, of a different shape.



$x^2 - y^2$: a minimum along one line, a maximum along another.

02

The second-derivative test

The idea

At a critical point $\mathbf{a}$ the linear term of Taylor's formula vanishes:

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \tfrac{1}{2}\,\mathbf{h}^{\top} H \mathbf{h} + o(\|\mathbf{h}\|^2), \qquad H = H_f(\mathbf{a}).$$

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Near $\mathbf{a}$, f looks like its quadratic term. By the spectral theorem (Linear Algebra, Unit 20) H has an orthonormal basis of eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$, and

$$\mathbf{h} = \sum_i c_i \mathbf{u}_i \quad \Longrightarrow \quad \mathbf{h}^{\top} H \mathbf{h} = \sum_i \lambda_i c_i^2.$$

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- All $\lambda_i > 0$: a bowl. All $\lambda_i < 0$: a cap. Mixed signs: a saddle.
- Some $\lambda_i = 0$: in those directions the quadratic term is flat, and the error term $o(\|\mathbf{h}\|^2)$ decides.

The second-derivative test

Theorem

Let f be C^2 near a critical point $\mathbf{a}$, and $H = H_f(\mathbf{a})$.

1. H positive definite (all $\lambda_i > 0$) $\Rightarrow$ $\mathbf{a}$ is a strict local minimum.
2. H negative definite (all $\lambda_i < 0$) $\Rightarrow$ $\mathbf{a}$ is a strict local maximum.
3. H indefinite (eigenvalues of both signs) $\Rightarrow$ $\mathbf{a}$ is a saddle point.

In every other case ($\det H = 0$, no eigenvalues of both signs) the test is **inconclusive**.

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Compare Calculus I: $f'(a) = 0$ and $f''(a) > 0$ give a minimum; $f''(a) = 0$ decides nothing (x^3, x^4, $-x^4$).

Proof

(1) Let $\lambda > 0$ be the smallest eigenvalue of H. Then $\mathbf{h}^\top H \mathbf{h} = \sum_i \lambda_i c_i^2 \geq \lambda \sum_i c_i^2 = \lambda \|\mathbf{h}\|^2$.

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$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \;\geq\; \tfrac{\lambda}{2}\|\mathbf{h}\|^2 - \tfrac{\lambda}{4}\|\mathbf{h}\|^2 \;=\; \tfrac{\lambda}{4}\|\mathbf{h}\|^2 \;>\; 0 \qquad (\mathbf{h} \neq \mathbf{0}).$$

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(2) Apply (1) to $-f$.

(3) Take unit eigenvectors $\mathbf{u}$, $\mathbf{v}$ with eigenvalues $\lambda > 0 > \mu$. Then

$$f(\mathbf{a}+t\mathbf{u}) - f(\mathbf{a}) = \tfrac{\lambda}{2}t^2 + o(t^2) > 0, \qquad f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a}) = \tfrac{\mu}{2}t^2 + o(t^2) < 0$$

for small $t \neq 0$: neither a minimum nor a maximum. □

Two variables: the discriminant

For $n = 2$, $H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}$ and $D = \det H = f_{xx} f_{yy} - f_{xy}^2 = \lambda_1 \lambda_2$.

The test for $f(x, y)$ at a critical point

$D > 0$ and $f_{xx} > 0$	local minimum
$D > 0$ and $f_{xx} < 0$	local maximum
$D < 0$	saddle point
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Why: $D < 0$ means λ_1, λ_2 have opposite signs. $D > 0$ means the same sign; then $f_{xx}f_{yy} > f_{xy}^2 \geq 0$, so f_{xx} and f_{yy} have that sign too, and so does $\lambda_1 + \lambda_2 = f_{xx} + f_{yy}$.

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In n variables, Sylvester's criterion (Linear Algebra, Unit 21): H is positive definite if and only if all its leading principal minors are positive.

Example: $f(x, y) = x^3 - 3x + y^2$

$\nabla f = (3x^2 - 3,\ 2y) = \mathbf{0}$ at $(1, 0)$ and $(-1, 0)$.

$$H = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}, \qquad D = 12x.$$

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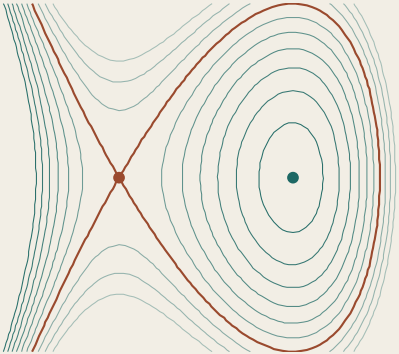
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The minimum is not global: $f(x, 0) = x^3 - 3x \to -\infty$ as $x \to -\infty$.



Level curves. The level $f = 2$ (brick) crosses itself at the saddle $(-1, 0)$; the minimum $(1, 0)$ sits inside closed curves.

Example: $f(x, y) = x^4 + y^4 - 4xy + 1$

$\nabla f = (4x^3 - 4y,\ 4y^3 - 4x) = \mathbf{0}$ gives $y = x^3$ and $x = y^3 = x^9$, so $x \in \{0, 1, -1\}$: the critical points are $(0, 0)$, $(1, 1)$, $(-1, -1)$.

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- $(0, 0)$: $D = -16 < 0$: saddle, $f = 1$.
- $(\pm 1, \pm 1)$: $D = 144 - 16 = 128 > 0$, $f_{xx} = 12 > 0$: local minima, $f = -1$.

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Global? $x^4 + y^4 \geq \frac{1}{2}(x^2 + y^2)^2$ and $4|xy| \leq 2(x^2 + y^2)$, so $f \to \infty$ as $\|(x, y)\| \to \infty$. Then f has a global minimum (extreme value theorem on a large disc), it is a critical point, and so the global minimum value is -1.

When the test is inconclusive

At $(0,0)$, these three functions all have $\nabla f = \mathbf{0}$ and $H = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, $D = 0$:

- $x^2 + y^4 \geq 0 = f(0,0)$: a strict minimum.
- $x^2 - y^4$: $f(t,0) = t^2 > 0$, $f(0,t) = -t^4 < 0$: a saddle.
- $x^2 + y^3$: $f(0,t) = t^3$ takes both signs: a saddle.

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> **$D = 0$ decides nothing**
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> The quadratic term is flat along the y-axis, and the higher-order terms decide. Look at f itself: along lines and curves through the point, or by completing squares.

03

Global extrema

Global extrema on a closed bounded set

Extreme value theorem (Lecture 9): a continuous function on a nonempty closed and bounded set $K \subseteq \mathbb{R}^n$ has a global minimum and a global maximum on K.

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Where are they? Either inside K — then, by Fermat, at a critical point — or on the boundary.

Recipe

1. Find the critical points in the interior of K.
2. Find the extreme values on the boundary: parametrise it (today), or use Lagrange multipliers (Lecture 16).
3. Compare all the values found. No second-derivative test is needed.

Example

Example ($f(x, y) = x^2 + y^2 - 2x$ on the disc $x^2 + y^2 \le 4$)

Interior. $\nabla f = (2x - 2,\ 2y) = \mathbf{0}$ at $(1, 0)$, where $f = -1$.

Check: $f = (x - 1)^2 + y^2 - 1$ is the squared distance from $(1, 0)$, minus 1. It is smallest at $(1, 0)$ and largest at the point of the disc farthest from $(1, 0)$.

Example

Example ($f(x,y) = x^2 + y^2 - 2x$ on the disc $x^2 + y^2 \le 4$)

Interior. $\nabla f = (2x - 2,\ 2y) = \mathbf{0}$ at $(1, 0)$, where $f = -1$.

Boundary. On $x^2 + y^2 = 4$: $f = 4 - 2x$ with $-2 \le x \le 2$. Smallest value 0 at $(2, 0)$, largest 8 at $(-2, 0)$.

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In one variable: if a differentiable $f : \mathbb{R} \to \mathbb{R}$ has only one critical point and it is a local minimum, it is the global minimum. **Not in two variables.**

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> $\nabla f = \big(2x - 3y^2(1 - x)^2,\ 2y(1 - x)^3\big)$. The second component vanishes if $y = 0$ or $x = 1$; then the first gives $x = 0$, or $2 = 0$. So $(0, 0)$ is the only critical point.

Moral: a local minimum becomes global only with an extra argument — growth at infinity, as for $x^4 + y^4 - 4xy + 1$, or convexity.

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> There $H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$: a strict local minimum, with $f = 0$.
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> But $f(4,1) = 16 - 27 = -11 < 0$.

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04

Convexity

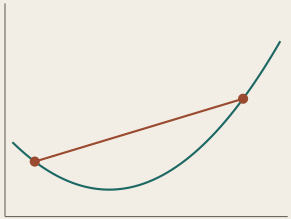
Convex sets and convex functions

Definition

- $C \subseteq \mathbb{R}^n$ is **convex** if it contains the segment between any two of its points.
- $f : C \to \mathbb{R}$ on a convex set is **convex** if for all $\mathbf{x}, \mathbf{y} \in C$ and $0 \le t \le 1$:

$$f\big((1-t)\mathbf{x} + t\mathbf{y}\big) \le (1-t)\,f(\mathbf{x}) + t\,f(\mathbf{y}).$$

- **Strictly convex:** $<$ whenever $\mathbf{x} \neq \mathbf{y}$ and $0 < t < 1$.
- f is **concave** if $-f$ is convex.



The chord lies above the graph. Along every line through C, f is convex.

Tests for convexity

Theorem

Let C be open and convex.

1. If f is C^1: f is convex $\iff$ $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in C$.
2. If f is C^2: f is convex $\iff$ $H_f(\mathbf{x})$ is positive semidefinite for every $\mathbf{x} \in C$. If $H_f(\mathbf{x})$ is positive definite for every $\mathbf{x}$, f is strictly convex.

- (1): the graph lies above every tangent plane.
- (2): the converse of the last statement fails — x^4 is strictly convex, but $f''(0) = 0$.

Why the tests work, and how to use them

Idea.

Restrict to lines: f is convex if and only if every $g(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ is, and $g''(t) = (\mathbf{y} - \mathbf{x})^{\top} H_f\,(\mathbf{y} - \mathbf{x})$ by the chain rule. Now use the one-variable facts (Calculus I, Unit 14). □

Why the tests work, and how to use them

Idea.

Restrict to lines: f is convex if and only if every $g(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ is, and $g''(t) = (\mathbf{y} - \mathbf{x})^\top H_f\,(\mathbf{y} - \mathbf{x})$ by the chain rule. Now use the one-variable facts (Calculus I, Unit 14). □

Examples.

- $x^2 + xy + y^2$: $H = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ at every point, with eigenvalues 3 and 1: strictly convex.
- $x^2 - y^2$: $H = \operatorname{diag}(2, -2)$ is indefinite: neither convex nor concave.
- $-\ln x - \ln y$ on $x, y > 0$: $H = \operatorname{diag}(1/x^2, 1/y^2)$: strictly convex.

Critical points of convex functions

Theorem

Let f be convex and C^1 on an open convex set C.

1. If $\nabla f(\mathbf{a}) = \mathbf{0}$, then $\mathbf{a}$ is a **global** minimum of f on C.
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(1) The tangent-plane inequality at $\mathbf{a}$: $f(\mathbf{y}) \geq f(\mathbf{a}) + \mathbf{0} \cdot (\mathbf{y} - \mathbf{a}) = f(\mathbf{a})$.

For convex functions, local = global: solving $\nabla f = \mathbf{0}$ is the whole problem.

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Examples of convex functions

- Affine functions $\mathbf{a} \cdot \mathbf{x} + c$: convex and concave ($H = 0$).
- $\mathbf{x}^\top Q \mathbf{x}$ with Q symmetric positive semidefinite ($H = 2Q$); $\|\mathbf{x}\|^2$ is strictly convex ($H = 2I$).
- Every norm, by the triangle inequality: $\|(1-t)\mathbf{x} + t\mathbf{y}\| \le (1-t)\|\mathbf{x}\| + t\|\mathbf{y}\|$. (Not differentiable at $\mathbf{0}$ — convexity does not need derivatives.)
- e^{x+y}: $H = e^{x+y}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, positive semidefinite: convex, but not strictly.

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Building convex functions

Sums of convex functions with weights ≥ 0, maxima of convex functions, and $\mathbf{x} \mapsto f(A\mathbf{x} + \mathbf{b})$ for convex f are convex.

Least squares, finished

From Lecture 11: $E(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$ has $\nabla E = 2A^\top A\mathbf{x} - 2A^\top \mathbf{b}$, and $\nabla E = \mathbf{0}$ are the normal equations $A^\top A\mathbf{x} = A^\top \mathbf{b}$.

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- If the columns of A are linearly independent, $A\mathbf{h} \neq \mathbf{0}$ for $\mathbf{h} \neq \mathbf{0}$: H_E is positive definite, E is strictly convex, and the minimiser $\mathbf{x} = (A^\top A)^{-1}A^\top\mathbf{b}$ is unique.

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> **Remark**
>
> Linear and logistic regression have convex losses; the losses of neural networks are not convex, and have many critical points. Next time: gradient descent, analysed on convex quadratics.

05

Summary and practice

Summary

- **Fermat**: at an interior local extremum of a differentiable f, $\nabla f = \mathbf{0}$. The converse fails: saddle points.
- **Second-derivative test** at a critical point: H positive definite $\Rightarrow$ minimum, negative definite $\Rightarrow$ maximum, indefinite $\Rightarrow$ saddle; otherwise inconclusive. For $n = 2$: the signs of $D = f_{xx} f_{yy} - f_{xy}^2$ and f_{xx}.
- **Global extrema** on a closed bounded set: interior critical points, the boundary, compare. A single local minimum need not be global.
- **Convex** f: chords above the graph; H_f positive semidefinite. Every critical point is a global minimum, and it is unique if f is strictly convex.
- Least squares is convex: the normal equations give the global minimum.

Practice

1. Find and classify the critical points of $x^3 + y^3 - 3xy$. Is the local minimum global?
2. Find the minimum of $x^2 + 4y^2 - 2x + 8y + 5$ on $\mathbb{R}^2$. Why is it global?
3. Find and classify the critical points of $xy\,(1 - x - y)$.
4. Show that the test is inconclusive for $x^2 + y^4$ and $x^2 - y^4$ at $(0, 0)$, and classify $(0, 0)$ for each.
5. Find the global maximum and minimum of $x^2 - y^2$ on the square $[-1, 1]^2$.
6. Show that $\|\mathbf{x}\|$ is convex on $\mathbb{R}^n$ but not strictly convex.
7. Show that $\|A\mathbf{x} - \mathbf{b}\|^2$ is strictly convex if the columns of A are linearly independent.

Answers

1. $(0,0)$: $D=-9$, saddle. $(1,1)$: $D=27$, $f_{xx}=6$, local minimum -1. Not global: $f(x,0)=x^3\to-\infty$.
2. $(x-1)^2+4(y+1)^2$: minimum 0 at $(1,-1)$. Global: f is convex ($H=\operatorname{diag}(2,8)$).
3. Saddles at $(0,0)$, $(1,0)$, $(0,1)$ ($D=-1$); local maximum $\frac{1}{27}$ at $(\frac{1}{3},\frac{1}{3})$ ($D=\frac{1}{3}$, $f_{xx}=-\frac{2}{3}$).
4. Both have $H=\operatorname{diag}(2,0)$, $D=0$. x^2+y^4: minimum; x^2-y^4: saddle.
5. Interior: a saddle at $(0,0)$. Boundary: $1-y^2$ on $x=\pm1$, x^2-1 on $y=\pm1$. Maximum 1 at $(\pm1,0)$, minimum -1 at $(0,\pm1)$.
6. Triangle inequality. Not strict: for $\mathbf{x}\neq\mathbf{0}$, $\|(1-t)\mathbf{x}+t\,(2\mathbf{x})\|=(1-t)\|\mathbf{x}\|+t\|2\mathbf{x}\|$.
7. $\mathbf{h}^\top H\mathbf{h}=2\|A\mathbf{h}\|^2>0$ for $\mathbf{h}\neq\mathbf{0}$, since $A\mathbf{h}\neq\mathbf{0}$.

Next time

Lecture 15 — Gradient descent (Unit 15)

- The algorithm: $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$
- Convergence for convex quadratics: the step size and the condition number
- Why it zigzags, and what can be done about it

Reading: Boyd & Vandenberghe, *Convex Optimization*, Sections 9.1–9.3.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.